

TD 4 — Duality

Lecture 5 · Exercise sheet

Exercise 1 Writing duals

(a) max $z = 3x_1 + 2x_2$ s.t. $x_1 + x_2 \le 4,\ x_1 + 3x_2 \le 9,\ x_1 \le 3,\ x_1, x_2 \ge 0$ (TD 2, Exercise 2).

(b) min $z = 6x_1 + 4x_2 + 3x_3$ s.t. $2x_1 + x_2 + x_3 \ge 20,\ x_1 + 2x_2 + x_3 \ge 30,\ x_1, x_2, x_3 \ge 0$.

(c) max $z = 2x_1 + x_2$ s.t. $x_1 + x_2 = 4,\ x_1 - x_2 \le 2,\ x_1 \ge 0,\ x_2$ free.

(d) min $z = x_1 + 2x_2$ s.t. $x_1 + x_2 \ge 2,\ x_1 - x_2 \le 1,\ x_1, x_2 \ge 0$.

1. Write the dual of each problem (Definition 5.1 or Rule 5.1). Give the sign of each dual variable.
2. In TD 2 we found that $x = (3,1)$ is optimal for (a), with $z = 11$. Check that $y = (2,0,1)$ is feasible for the dual of (a), and use it to prove again that $x = (3,1)$ is optimal.
3. The optimal plans of (c) and (d) are $x = (3,1)$ and $x = (\frac{3}{2}, \frac{1}{2})$. Check that $y = (\frac{3}{2}, \frac{1}{2})$ and $y = (\frac{3}{2}, -\frac{1}{2})$ are feasible for the duals of (c) and (d), and that they give the same values as the plans.

Exercise 2 Control boards: the price of a minute

Recall the control boards of TD 1 and TD 2: x_1, x_2 = numbers of Basic and Pro boards per day (profit in thousand DA),

$$\max\ z = 12x_1 + 20x_2 \qquad \text{s.t.} \qquad \begin{cases} 2x_1 + 4x_2 \le 480 & \text{(assembly machine, minutes)} \\ 3x_1 + 2x_2 \le 360 & \text{(test bench, minutes)} \\ x_2 \le 100 & \text{(special sensors)} \\ x_1, x_2 \ge 0. & \end{cases}$$

In TD 2 the last dictionary was $z = 2520 - \frac{9}{2}s_1 - s_2$, with the optimal plan $(60, 90)$.

1. Write the dual. Give the meaning of each dual variable and of each dual constraint.
2. Read the optimal dual solution in the last z-line (Key fact 5.3). Check that it is dual feasible and that $w = z^*$.
3. Guess the new best profit with 481 machine minutes, then with 361 bench minutes, then with 101 sensors. Check the first two guesses by solving the two equations of the binding constraints.
4. Use the shadow prices to decide (check each answer by solving the two equations again):
   (a) A partner rents 20 extra machine minutes per day at 4 thousand DA per minute. Accept?
   (b) Another partner wants to buy 10 bench minutes per day at 1.5 thousand DA per minute. Sell?
5. A new board "Mini" needs 1 machine minute, 2 bench minutes and no sensor. For which profits is it worth making? Is it worth making with a profit of 6? Of 7?

Exercise 3 Three robot arms: complementary slackness

A workshop makes three robot arms P1, P2, P3. Each week it has 18 servo motors, 30 assembly hours and 14 calibration hours. With x_j the number of arms Pj per week (profit in thousand DA):

$$\max\ z = 3x_1 + 5x_2 + 6x_3 \qquad \text{s.t.} \qquad \begin{cases} x_2 + x_3 \le 18 & \text{(servo motors)} \\ 2x_1 + 3x_2 + 2x_3 \le 30 & \text{(assembly hours)} \\ 4x_2 + 2x_3 \le 14 & \text{(calibration hours)} \\ x_1, x_2, x_3 \ge 0. \end{cases}$$

1. Write the dual.
2. A manager proposes the plan $x = (0, 0, 7)$. Check that it is feasible, and use Method 5.1 to show that it is *not* optimal. Explain the result in words.
3. Use Method 5.1 to show that $x = (8, 0, 7)$ is optimal. Give the dual solution and check that $w = z$.
4. Why is P2 not made? From which profit would P2 become interesting?

Exercise 4 Charging a battery: a negative price

Recall the battery of TD 1 (Exercise 3) and TD 3 (Exercise 2): t_F and t_S are the hours on the fast and on the slow charger,

$$\min\ z = 300\,t_F + 80\,t_S \quad \text{s.t.} \quad 2t_F + t_S \ge 24\ \text{(energy, kWh)}, \quad t_F + t_S \le 16\ \text{(time, hours)}, \quad t_F, t_S \ge 0.$$

The cheapest plan is $(t_F, t_S) = (8, 8)$, with cost 3040.

1. Write the dual with Rule 5.1. What are the signs of y_1 and y_2?
2. Use complementary slackness and the plan $(8, 8)$ to find the dual solution. Check it.
3. Explain the meaning of y_1 and y_2. Why is it normal that y_2 is negative? Compare with the numbers 220 and 140 of the last tableau of TD 3.
4. Draw the feasible set of the dual (with $y_1 \ge 0$, $y_2 \le 0$) and find its optimum by drawing.

Exercise 5 At home — Robot kits: from the dual to the primal

A robotics club needs at least 20 motors and 30 sensors. It can buy three kinds of kits: kit 1 (2 motors, 1 sensor) for 6 thousand DA, kit 2 (1 motor, 2 sensors) for 4 thousand DA, kit 3 (1 motor, 1 sensor) for 3 thousand DA. This is problem (b) of Exercise 1.

1. Use the dual of Exercise 1(b). Give the meaning of y_1 and y_2, and of each dual constraint.
2. The dual has two variables: solve it by drawing (give all the corners and their values).
3. Use complementary slackness to find the cheapest purchase from the dual solution. Check its cost.
4. Explain with the prices why kit 1 is not bought.

Exercise 6 At home — All the cases

1. In TD 2 (Exercise 3a) the problem max $z = x_1 + x_2$ s.t. $-x_1 + x_2 \leq 2$, $x_1 - 2x_2 \leq 2$, $x_1, x_2 \geq 0$ was unbounded. Write its dual, and show directly that the dual has no feasible solution. Which key fact predicted it?
2. True or false? Justify each answer.
   (a) If the primal has no feasible solution, the dual is unbounded.
   (b) If x is primal feasible, y is dual feasible and $c^\top x = b^\top y = 100$, then x and y are both optimal.
   (c) A maximisation can have a feasible plan with $z = 50$ while its dual has a feasible solution with $w = 40$.
   (d) If a resource is not fully used in an optimal plan, its shadow price is 0.
   (e) If the optimal price y_i of a resource is 0, then this resource is not fully used.